Development of the Periodic Table

AQA GCSE Combined Science: Trilogy · Chemistry Topic 1 · Revision activity 4 · 16 marks · about 25 minutes

Read the key information (5 min), answer the **quick recall** questions without looking back (5 min), then answer the **exam questions** (15 min). Mark your work with the mark scheme on the last page.

Key information

- Before protons, neutrons and electrons were discovered, scientists ordered the elements by their **atomic weight**.
- Early periodic tables were **incomplete**, and some elements were put in the **wrong groups** because the order of atomic weights was followed strictly.
- **Mendeleev (1869)** also ordered elements by atomic weight, but he:
  - **swapped the order** of some elements so that elements with similar properties were in the same group (e.g. tellurium and iodine)
  - **left gaps** for elements that had not been discovered yet, and **predicted** their properties.
- When elements such as gallium and **germanium** were discovered, their properties matched Mendeleev's predictions. This made other scientists accept his table.
- Later, the discovery of **isotopes** explained why ordering by atomic weight was not always correct. The modern periodic table is ordered by **atomic number**.

Mendeleev's prediction for 'eka-silicon' compared with germanium (discovered 1886)

Property	Mendeleev's prediction (1871)	Germanium (actual)
Relative atomic mass	72	72.6
Density in g/cm^3	5.5	5.3
Colour	grey	grey-white
Formula of oxide	XO_2	GeO_2

Mendeleev's table compared with the modern periodic table

	Mendeleev's table	Modern periodic table
Elements ordered by	atomic weight (with some swapped)	atomic number
Gaps?	yes – for undiscovered elements	no gaps
Noble gases (group 0)	not included – not yet discovered	included
Number of elements	about 60	118

Section A: Quick recall (6 marks – 1 mark each)

1 Before atomic structure was known, how did scientists order the elements?

2 Which scientist left gaps in his periodic table?

3 Why did Mendeleev leave gaps?

4 How are the elements ordered in the modern periodic table?

5 Which group of elements was missing from Mendeleev's table?

6 Name one element whose properties Mendeleev predicted.

Section B: Exam questions

01 Give two ways in which Mendeleev's periodic table was different from earlier periodic tables.

[2 marks]

02 Explain why other scientists came to accept Mendeleev's periodic table.

[2 marks]

03 Tellurium has an atomic weight of 128. Iodine has an atomic weight of 127. Mendeleev placed iodine **after** tellurium. Explain why.

[2 marks]

04 Use the eka-silicon table to explain whether Mendeleev's prediction was a good one.

[2 marks]

05 Explain how the discovery of isotopes showed that ordering elements by atomic weight is not always correct.

[2 marks]

Mark Scheme – Development of the Periodic Table

(1) = one mark. / separates acceptable alternatives. Words in brackets are not needed for the mark.

Question	Marking guidance	Comments	Mark
A	**1** by atomic weight **2** Mendeleev **3** for elements that had not been discovered yet **4** by atomic number **5** the noble gases (group 0) **6** germanium (or gallium)	1 mark each	6
01	he left gaps for undiscovered elements (1) he changed the order of some elements so that elements with similar properties were in the same group (1)		2
02	elements were discovered that filled the gaps (1) their properties matched Mendeleev's predictions (1)		2
03	iodine has similar properties to chlorine and bromine (1) so it needed to be in the same group as them (group 7) – properties are more important than strict atomic weight order (1)		2
04	yes, it was a good prediction (1) e.g. relative atomic mass 72 v 72.6 / density 5.5 v 5.3 / same oxide formula XO_2 – values are very close (1)		2
05	isotopes have different numbers of neutrons, which changes the atomic weight but not the number of protons (1) an element can have a higher atomic weight than the next element even though it has fewer protons (e.g. tellurium and iodine) (1)		2

Total: 16 marks